

Wenyuan Wang

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RESEARCH INTERESTS

My research focuses on multimodal learning, world models and interpretability. I aim to develop AI systems capable of robust causal forecasting and grounded understanding of the physical world.

EDUCATION

Wuhan University

B.S. in Electronic Information Engineering

Sep 2020 - June 2024

– GPA: 3.60(87.2)/4.0

Publications

• Physics Plot

Wenyuan Wang*, Wentao Li*, Kuei-Chun Kao, Rui Pan, Tong Zhang

ARR, ACL2026 (Submitted to ACL)

• Superstructuring a Task-Sufficient World Model on Top of Visual Foundational Model

Minghao Fu, Fan Feng, Wenyuan Wang, Biwei Huang

WMW 2026 Workshop (Under Review)

- Proposed to analyze the latent space of the world model using t-SNE clustering visualization and to further interpret our model by visualizing its attention patterns.

• MMMG: A Comprehensive and Reliable Benchmark for Multitask Multimodal Generation

Jihan Yao, *Yushi Hu*, Wenyuan Wang, Bin Han, Shangbin Feng, Guang Yang, Yujie Yi, Bingbing Wen, Ranjay Krishna, Lucy Lu Wang, Yulia Tsvetkov, Noah A. Smith, Banghua Zhu

ICLR 2026 (Under Review)

- Extended the benchmark with additional examples, Evaluated openai sourced generative models about t2i tasks and performed a systematic, in-depth analysis of failure cases.

• Generalizable Geometric Image Caption Synthesis

Yue Xin*, Wenyuan Wang*, Rui Pan, Ruida Wang, Howard Meng, Renjie Pi, Shizhe Diao, Tong Zhang

arXiv:2509.15217/ICLR 2026 (Under Review)

- Integrated symbolic reasoning with neural-guided heuristics, constructed a verifiable pipeline for image-pair generation.

• Probabilistic Residual User Clustering

Wenyuan Wang, Yusong Zhao, Zihao Xu, Hengyi Wang, Shreya Venugopal, Desmond Lobo, Chengzhi Mao,

Qi Xu, Zhigang Hua, Yan Xie, Bo Long, Shuang Yang, Hao Wang

IJCAI2025 workshop / Submitted to TMLR

- Proposed PRUC, a causal Bayesian framework that clusters users and models residuals to enhance recommendation accuracy
- Introduced plug-and-play architecture compatible with diverse deep learning recommenders, improving cold-start performance

• Multimodal Needle in a Haystack: Benchmarking Long-Context Capability of MLLMs

Hengyi Wang, Haizhou Shi, Shiwei Tan, Weiye Qin, Wenyuan Wang, Tunyu Zhang, Akshay Nambi, Tanuja

Ganu, Hao Wang

NAACL 2025 Main

- Evaluated performance of InstructBLIP on multimodal long-context benchmark
- Contributed to comprehensive evaluation methodology for multimodal LLM capabilities

• Continual Learning of Large Language Models: A Comprehensive Survey

Haizhou Shi, Zihao Xu, Hengyi Wang, Weiyi Qin, **Wenyuan Wang**, Yibin Wang, Zifeng Wang, Sayna Ebrahimi, Hao Wang
ACM Computing Surveys

- Researched and documented advancements in continual learning for multimodal LLMs
- Contributed to comprehensive survey covering state-of-the-art techniques in the field

• **Multi-tailed Vision Transformer for Efficient Inference**

Yunke Wang, Bo Du, **Wenyuan Wang**, Chang Xu
Neural Networks, 2024, 174: 106235

- Designed multiple tails to generate visual sequences of different lengths for Transformer processing
- Achieved significant FLOP reduction with no accuracy degradation compared to baseline methods

RESEARCH EXPERIENCE

Physics Lean Prover

Advisor: Tong Zhang

- curated dataset containing 3,122 training examples from the PhysLean library, covering topics such as category theory, tensor algebra, and relativity physics. Finetuned Goedel-Prover-V2 on this domain-specific dataset.
- Proposed an evaluation framework utilizing Lean server verification, enabling scalable and reliable assessment of theorem proving in physics

Steering Vector Analysis in Large Reasoning Models

Advisor: Canyu Chen

- Investigated how steering-vector-based interventions affect the robustness and faithfulness of reasoning in large reasoning models.
- Analyzed intrinsic properties of steering vectors and demonstrated how specific latent directions systematically induce, modify, or bias step-by-step reasoning trajectories.

Interpretability of Multimodal Large Language Models

Advisor: Prof. Hao Wang

- Identified strong attention correlations between visual inputs and token-level outputs in LLaVA.
- Proposed an adaptive pruning technique for hierarchical attention layers, improving both accuracy and inference efficiency.

Dynamic Scene Graph Generation

Advisor: Prof. Runhao Zeng, Prof. Chuang Gan

- Deployed a MiniGPT-4-based multimodal LLM for scene graph generation with joint reasoning over class, box, and relationship triples.
- Fine-tuned class and bounding-box models via transfer learning to improve relationship prediction accuracy.

Dynamic Confidence Calibration with Adaptive Temperature Adjustment

Advisor: Prof. Chang Xu

- Proposed a dynamic neural network that allocates computational resources according to sample difficulty.
- Applied adaptive temperature adjustment to improve confidence calibration and sample difficulty verification.

WORK EXPERIENCE

Visual Alignment in Industrial Settings

Aug 2023 – Jan 2024

Intern, Siemens

Shanghai, China

- Simulated AGV-based cargo detection in Gazebo using ROS2.
- Designed a vision algorithm to estimate cargo pose and position by analyzing frontal-area geometry.

SKILLS

- **Programming:** Python, C++, C, Lean, Pytorch
- **Tools & Platforms:** ROS, Linux, Docker, Embedded Systems, Gazebo, IsaacSim